6.2 The Directions

These directions are similar to those you will see for data sufficiency questions when you take the GMAT test. If you read the directions carefully and understand them clearly before going to sit for the test, you will not need to spend much time reviewing them when you take the GMAT test.

Each data sufficiency problem consists of a question and two statements, labeled (1) and (2), that give data. You have to decide whether the data given in the statements are *sufficient* for answering the question. Using the data given in the statements *plus* your knowledge of mathematics and everyday facts (such as the number of days in July or the meaning of *counterclockwise*), you must indicate whether the data given in the statements are sufficient for answering the questions and then indicate one of the following answer choices:

(A) Statement (1) ALONE is sufficient, but statement (2) alone is not sufficient to answer the question asked;

(B) Statement (2) ALONE is sufficient, but statement (1) alone is not sufficient to answer the question asked;

(C) BOTH statements (1) and (2) TOGETHER are sufficient to answer the question asked, but NEITHER statement ALONE is sufficient;

(D) EACH statement ALONE is sufficient to answer the question asked;

(E) Statements (1) and (2) TOGETHER are NOT sufficient to answer the question asked, and additional data are needed.

NOTE: In data sufficiency problems that ask for the value of a quantity, the data given in the statements are sufficient only when it is possible to determine exactly one numerical value for the quantity.

Numbers: All numbers used are real numbers.

Figures: A figure accompanying a data sufficiency problem will conform to the information given in the question but will not necessarily conform to the additional information given in statements (1) and (2).

Lines shown as straight can be assumed to be straight and lines that appear jagged can also be assumed to be straight.

You may assume that the positions of points, angles, regions, and so forth exist in the order shown and that angle measures are greater than zero degrees.

All figures lie in a plane unless otherwise indicated.

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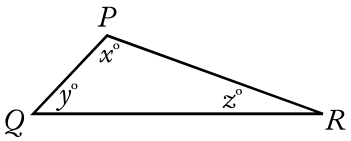
6.3 Sample Questions

Each data sufficiency problem consists of a question and two statements, labeled (1) and (2), which contain certain data. Using these data and your knowledge of mathematics and everyday facts (such as the number of days in July or the meaning of the word *counterclockwise*), decide whether the data given are sufficient for answering the question and then indicate one of the following answer choices:

A Statement (1) ALONE is sufficient, but statement (2) alone is not sufficient.
B Statement (2) ALONE is sufficient, but statement (1) alone is not sufficient.
C BOTH statements TOGETHER are sufficient, but NEITHER statement ALONE is sufficient.
D EACH statement ALONE is sufficient.
E Statements (1) and (2) TOGETHER are not sufficient.

Note: In data sufficiency problems that ask for the value of a quantity, the data given in the statements are sufficient only when it is possible to determine exactly one numerical value for quantity.

Example:



In $\triangle PQR$, what is the value of x?

(1) $PQ = PR$
(2) $y = 40$

Explanation: According to statement (1) $PQ = PR$; therefore, $\triangle PQR$ is isosceles and $y = z$. Since $x + y + z = 180$, it follows that $x + 2y = 180$. Since statement (1) does not give a value for y, you cannot answer the question using statement (1) alone. According to statement (2), $y = 40$; therefore, $x + z = 140$. Since statement (2) does not give a value for z, you cannot answer the question using statement (2) alone. Using both statements together, since $x + 2y = 180$ and the value of y is given, you can find the value of x. Therefore, BOTH statements (1) and (2) TOGETHER are sufficient to answer the questions, but NEITHER statement ALONE is sufficient.

Numbers: All numbers used are real numbers.

Figures:

- Figures conform to the information given in the question, but will not necessarily conform to the additional information given in statements (1) and (2).
- Lines shown as straight are straight, and lines that appear jagged are also straight.
- The positions of points, angles, regions, etc., exist in the order shown, and angle measures are greater than zero.
- All figures lie in a plane unless otherwise indicated.

1. What is the value of $|x|$?

   (1) $x = -|x|$
   (2) $x^2 = 4$

2. What percent of a group of people are women with red hair?

   (1) Of the women in the group, 5 percent have red hair.
   (2) Of the men in the group, 10 percent have red hair.

3. In a certain class, one student is to be selected at random to read. What is the probability that a boy will read?

   (1) Two-thirds of the students in the class are boys.
   (2) Ten of the students in the class are girls.

4. In College X the number of students enrolled in both a chemistry course and a biology course is how much less than the number of students enrolled in neither?

   (1) In College X there are 60 students enrolled in a chemistry course.
   (2) In College X there are 85 students enrolled in a biology course.

5. A certain expressway has Exits J, K, L, and M, in that order. What is the road distance from Exit K to Exit L ?

   (1) The road distance from Exit J to Exit L is 21 kilometers.
   (2) The road distance from Exit K to Exit M is 26 kilometers.

6. If n is an integer, is $n + 1$ odd?

   (1) $n + 2$ is an even integer.
   (2) $n - 1$ is an odd integer.

7. For which type of investment, J or K, is the annual rate of return greater?

   (1) Type J returns \$115 per \$1,000 invested for any one-year period and type K returns \$300 per \$2,500 invested for any one-year period.
   (2) The annual rate of return for an investment of type K is 12 percent.

8. A citrus fruit grower receives \$15 for each crate of oranges shipped and \$18 for each crate of grapefruit shipped. How many crates of oranges did the grower ship last week?

   (1) Last week the number of crates of oranges that the grower shipped was 20 more than twice the number of crates of grapefruit shipped.
   (2) Last week the grower received a total of \$38,700 from the crates of oranges and grapefruit shipped.

9. If Pat saved \$600 of his earnings last month, how much did Pat earn last month?

   (1) Pat spent $\frac{1}{2}$ of his earnings last month for living expenses and saved $\frac{1}{3}$ of the remainder.
   (2) Of his earnings last month, Pat paid twice as much in taxes as he saved.

10. Water is pumped into a partially filled tank at a constant rate through an inlet pipe. At the same time, water is pumped out of the tank at a constant rate through an outlet pipe. At what rate, in gallons per minute, is the amount of water in the tank increasing?

    (1) The amount of water initially in the tank is 200 gallons.
    (2) Water is pumped into the tank at a rate of 10 gallons per minute and out of the tank at a rate of 10 gallons every $2\frac{1}{2}$ minutes.

11. Is x a negative number?

    (1) $9x > 10x$
    (2) $x + 3$ is positive.

12. If i and j are integers, is $i + j$ an even integer?

    (1) $i < 10$
    (2) $i = j$

13. The charge for a telephone call between City R and City S is \$0.42 for each of the first 3 minutes and \$0.18 for each additional minute. A certain call between these two cities lasted for x minutes, where x is an integer. How many minutes long was the call?

    (1) The charge for the first 3 minutes of the call was \$0.36 less than the charge for the remainder of the call.
    (2) The total charge for the call was \$2.88.

14. If Car X followed Car Y across a certain bridge that is $\frac{1}{2}$ mile long, how many seconds did it take Car X to travel across the bridge?

(1) Car X drove onto the bridge exactly 3 seconds after Car Y drove onto the bridge and drove off the bridge exactly 2 seconds after Car Y drove off the bridge.

(2) Car Y traveled across the bridge at a constant speed of 30 miles per hour.

15. If $n + k = m$, what is the value of k?

(1) $n = 10$

(2) $m + 10 = n$

16. Is x an integer?

(1) $\frac{x}{2}$ is an integer.

(2) $2x$ is an integer.

17. Is the integer P odd?

(1) The sum of P, $P + 4$, and $P + 11$ is even.

(2) The sum of $P - 3$, P, and $P + 11$ is odd.

18. What is the maximum number of rectangular blocks, each with dimensions 12 centimeters by 6 centimeters by 4 centimeters, that will fit inside rectangular Box X?

(1) When Box X is filled with the blocks and rests on a certain side, there are 25 blocks in the bottom layer.

(2) The inside dimensions of Box X are 60 centimeters by 30 centimeters by 20 centimeters.

19. If sequence S has 200 terms, what is the 192nd term of S?

(1) The first term of S is -40.

(2) Each term of S after the first term is 3 less than the preceding term.

20. In ΔPQR, if $PQ = x$, $QR = x + 2$, and $PR = y$, which of the three angles of ΔPQR has the greatest degree measure?

(1) $y = x + 3$

(2) $x = 2$

21. What percent of the drama club members enrolled at a certain school are female students?

(1) Of the female students enrolled at the school, 40 percent are members of the drama club.

(2) Of the male students enrolled at the school, 25 percent are members of the drama club.

22. A family-size box of cereal contains more cereal and costs more than the regular-size box of cereal. What is the cost per ounce of the family-size box of cereal?

(1) The family-size box of cereal contains 10 ounces more than the regular-size box of cereal.

(2) The family-size box of cereal costs $5.40.

23. The profit from the sale of a certain appliance increases, though not proportionally, with the number of units sold. Did the profit exceed $4 million on sales of 380,000 units?

(1) The profit exceeded $2 million on sales of 200,000 units.

(2) The profit exceeded $5 million on sales of 350,000 units.

24. If n is an integer, is n even?

(1) $n^2 - 1$ is an odd integer.

(2) $3n + 4$ is an even integer.

25. Carmen currently works 30 hours per week at her part-time job. If her gross hourly wage were to increase by $1.50, how many fewer hours could she work per week and still earn the same gross weekly pay as before the increase?

(1) Her gross weekly pay is currently $225.00.

(2) An increase of $1.50 would represent an increase of 20 percent of her current gross hourly wage.

26. The number n of units of its product that Company X is scheduled to produce in month t of its next fiscal year is given by the formula $n = \frac{900}{1 + c2^{-t}}$, where c is a constant and t is a positive integer between 1 and 6, inclusive. What is the number of units of its product that Company X is scheduled to produce in month 6 of its next fiscal year?

(1) Company X is scheduled to produce 180 units of its product in month 1 of its next fiscal year.

(2) Company X is scheduled to produce 300 units of its product in month 2 of its next fiscal year.

27. When 200 gallons of oil were removed from a tank, the volume of oil left in the tank was $\frac{3}{7}$ of the tank's capacity. What was the tank's capacity?

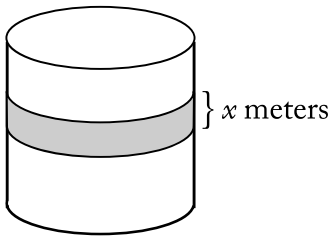
(1) Before the 200 gallons were removed, the volume of oil in the tank was $\frac{1}{2}$ of the tank's capacity.

(2) After the 200 gallons were removed, the volume of the oil left in the tank was 1,600 gallons less than the tank's capacity.

28. Division R of Company Q has 1,000 employees. What is the average (arithmetic mean) annual salary of the employees at Company Q ?

(1) The average annual salary of the employees in Division R is $30,000.

(2) The average annual salary of the employees at Company Q who are not in Division R is $35,000.



29. A circular tub has a band painted around its circumference, as shown above. What is the surface area of this painted band?

(1) $x = 0.5$

(2) The height of the tub is 1 meter.

30. What is the value of integer n ?

(1) $n(n + 1) = 6$

(2) $2^{2n} = 16$

$$d = 0.43t7$$

31. If t denotes the thousandths digit in the decimal representation of d above, what digit is t ?

(1) If d were rounded to the nearest hundredth, the result would be 0.44.

(2) If d were rounded to the nearest thousandth, the result would be 0.436.

32. Jerry bought 7 clothing items, including a coat, and the sum of the prices of these items was $365. If there was no sales tax on any clothing item with a price of less than $100 and a 7 percent sales tax on all other clothing items, what was the total sales tax on the 7 items that Jerry bought?

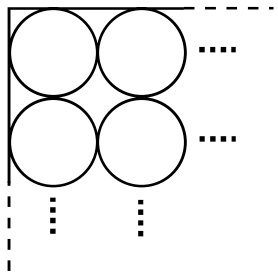
(1) The price of the coat was $125.

(2) The average (arithmetic mean) price for the 6 items other than the coat was $40.

33. What was the price at which a merchant sold a certain appliance?

(1) The merchant's gross profit on the appliance was 20 percent of the price at which the merchant sold the appliance.

(2) The price at which the merchant sold the appliance was $50 more than the merchant's cost of the appliance.



34. The inside of a rectangular carton is 48 centimeters long, 32 centimeters wide, and 15 centimeters high. The carton is filled to capacity with k identical cylindrical cans of fruit that stand upright in rows and columns, as indicated in the figure above. If the cans are 15 centimeters high, what is the value of k ?

(1) Each of the cans has a radius of 4 centimeters.

(2) Six of the cans fit exactly along the length of the carton.

$$\begin{cases} x - 4 = z \\ y - x = 8 \\ 8 - z = t \end{cases}$$

35. For the system of equations given, what is the value of z ?

(1) $x = 7$

(2) $t = 5$

36. For all integers n, the function f is defined by $f(n) = a^n$, where a is a constant. What is the value of $f(1)$?

(1) $f(2) = 100$
(2) $f(3) = -1{,}000$

37. The selling price of an article is equal to the cost of the article plus the markup. The markup on a certain television set is what percent of the selling price?

(1) The markup on the television set is 25 percent of the cost.
(2) The selling price of the television set is $250.

38. If p_1 and p_2 are the populations and r_1 and r_2 are the numbers of representatives of District 1 and District 2, respectively, the ratio of the population to the number of representatives is greater for which of the two districts?

(1) $p_1 > p_2$
(2) $r_2 > r_1$

39. In a random sample of 80 adults, how many are college graduates?

(1) In the sample, the number of adults who are not college graduates is 3 times the number who are college graduates.
(2) In the sample, the number of adults who are not college graduates is 40 more than the number who are college graduates.

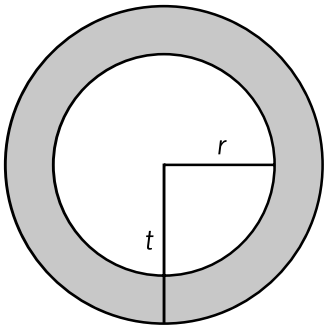
	R	S	T	U
R	0	y	x	62
S	y	0	56	75
T	x	56	0	69
U	62	75	69	0

40. The table above shows the distance, in kilometers, by the most direct route, between any two of the four cities, R, S, T, and U. For example, the distance between City R and City U is 62 kilometers. What is the value of x?

(1) By the most direct route, the distance between S and T is twice the distance between S and R.
(2) By the most direct route, the distance between T and U is 1.5 times the distance between R and T.

41. What is the value of the two-digit integer x?

(1) The sum of the two digits is 3.
(2) x is divisible by 3.



42. The figure above shows the circular cross section of a concrete water pipe. If the inside radius of the pipe is r feet and the outside radius of the pipe is t feet, what is the value of r?

(1) The ratio of $t - r$ to r is 0.15 and $t - r$ is equal to 0.3 foot.
(2) The area of the concrete in the cross section is 1.29π square feet.

43. What is the tenths digit in the decimal representation of a certain number?

(1) The number is less than $\frac{1}{3}$.
(2) The number is greater than $\frac{1}{4}$.

44. Robots X, Y, and Z each assemble components at their respective constant rates. If r_x is the ratio of Robot X's constant rate to Robot Z's constant rate and r_y is the ratio of Robot Y's constant rate to Robot Z's constant rate, is Robot Z's constant rate the greatest of the three?

(1) $r_x < r_y$
(2) $r_y < 1$

45. If r is a constant and $a_n = rn$ for all positive integers n, for how many values of n is $a_n < 100$?

(1) $a_{50} = 500$
(2) $a_{100} + a_{105} = 2{,}050$

46. If r is represented by the decimal $0.t5$, what is the digit t?

(1) $r < \frac{1}{3}$

(2) $r < \frac{1}{10}$

47. If the two floors in a certain building are 9 feet apart, how many steps are there in a set of stairs that extends from the first floor to the second floor of the building?

(1) Each step is $\frac{3}{4}$ foot high.

(2) Each step is 1 foot wide.

48. In June 1989, what was the ratio of the number of sales transactions made by Salesperson X to the number of sales transactions made by Salesperson Y?

(1) In June 1989, Salesperson X made 50 percent more sales transactions than Salesperson Y did in May 1989.

(2) In June 1989, Salesperson Y made 25 percent more sales transactions than in May 1989.

49. If $a < x < b$ and $c < y < d$, is $x < y$?

(1) $a < c$

(2) $b < c$

50. How many people are directors of both Company K and Company R?

(1) There were 17 directors present at a joint meeting of the directors of Company K and Company R, and no directors were absent.

(2) Company K has 12 directors and Company R has 8 directors.

51. If x and y are positive, is $\frac{x}{y}$ greater than 1?

(1) $xy > 1$

(2) $x - y > 0$

52. A clothing store acquired an item at a cost of x dollars and sold the item for y dollars. The store's gross profit from the item was what percent of its cost for the item?

(1) $y - x = 20$

(2) $\frac{y}{x} = \frac{5}{4}$

$$(n - x) + (n - y) + (n - z) + (n - k)$$

53. What is the value of the expression above?

(1) The average (arithmetic mean) of x, y, z, and k is n.

(2) x, y, z, and k are consecutive integers.

54. A taxi company charges f cents for the first mile of the taxi ride and m cents for each additional mile. How much does the company charge for a 10-mile taxi ride?

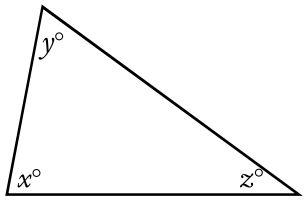
(1) The company charges \$0.90 for a 2-mile ride.

(2) The company charges \$1.20 for a 4-mile ride.

55. Guy's net income equals his gross income minus his deductions. By what percent did Guy's net income change on January 1, 1989, when both his gross income and his deductions increased?

(1) Guy's gross income increased by 4 percent on January 1, 1989.

(2) Guy's deductions increased by 15 percent on January 1, 1989.



56. What is the value of z in the triangle above?

(1) $x + y = 139$

(2) $y + z = 108$

57. Max has \$125 consisting of bills each worth either \$5 or \$20. How many bills worth \$5 does Max have?

(1) Max has fewer than 5 bills worth \$5 each.

(2) Max has more than 5 bills worth \$20 each.

58. If the ratio of the number of teachers to the number of students is the same in School District M and School District P, what is the ratio of the number of students in School District M to the number of students in School District P?

(1) There are 10,000 more students in School District M than there are in School District P.

(2) The ratio of the number of teachers to the number of students in School District M is 1 to 20.

59. If a total of 84 students are enrolled in two sections of a calculus course, how many of the 84 students are female?

(1) $\frac{2}{3}$ of the students in Section 1 are female.

(2) $\frac{1}{2}$ of the students in Section 2 are male.

60. What is the value of n in the equation $-25 + 19 + n = s$?

(1) $s = 2$

(2) $\frac{n}{s} = 4$

61. At a certain picnic, each of the guests was served either a single scoop or a double scoop of ice cream. How many of the guests were served a double scoop of ice cream?

(1) At the picnic, 60 percent of the guests were served a double scoop of ice cream.

(2) A total of 120 scoops of ice cream were served to all the guests at the picnic.

62. For a convention, a hotel charges a daily room rate of \$120 for 1 person and x dollars for each additional person. What is the charge for each additional person?

(1) The daily cost per person for 4 people sharing the cost of a room equally is \$45.

(2) The daily cost per person for 2 people sharing the cost of a room equally is \$25 more than the corresponding cost for 4 people.

63. Stores L and M each sell a certain product at a different regular price. If both stores discount their regular price of the product, is the discount price at Store M less than the discount price at Store L?

(1) At Store L the discount price is 10 percent less than the regular price; at Store M the discount price is 15 percent less than the regular price.

(2) At Store L the discount price is \$5 less than the regular store price; at Store M the discount price is \$6 less than the regular price.

64. If d denotes a decimal, is $d \geq 0.5$?

(1) When d is rounded to the nearest tenth, the result is 0.5.

(2) When d is rounded to the nearest integer, the result is 1.

65. How many integers are there between, but not including, integers r and s?

(1) $s - r = 10$

(2) There are 9 integers between, but not including, $r + 1$ and $s + 1$.

66. If n and t are positive integers, is n a factor of t?

(1) $n = 3^{n-2}$

(2) $t = 3^n$

67. In a survey of 200 college graduates, 30 percent said they had received student loans during their college careers, and 40 percent said they had received scholarships. What percent of those surveyed said that they had received neither student loans nor scholarships during their college careers?

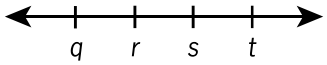
(1) 25 percent of those surveyed said that they had received scholarships but no loans.

(2) 50 percent of those surveyed who said that they had received loans also said that they had received scholarships.

68. Three machines, K, M, and P, working simultaneously and independently at their respective constant rates, can complete a certain task in 24 minutes. How long does it take Machine K, working alone at its constant rate, to complete the task?

(1) Machines M and P, working simultaneously and independently at their respective constant rates, can complete the task in 36 minutes.

(2) Machines K and P, working simultaneously and independently at their respective constant rates, can complete the task in 48 minutes.



69. Of the four numbers represented on the number line above, is r closest to zero?

(1) $q = -s$

(2) $-t < q$

70. Mary persuaded n friends to donate \$500 each to her election campaign, and then each of these n friends persuaded n more people to donate \$500 each to Mary's campaign. If no one donated more than once and if there were no other donations, what was the value of n?

(1) The first n people donated $\frac{1}{16}$ of the total amount donated.
(2) The total amount donated was \$120,000.

71. Carlotta can drive from her home to her office by one of two possible routes. If she must also return by one of these routes, what is the distance of the shorter route?

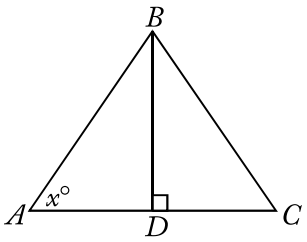
(1) When she drives from her home to her office by the shorter route and returns by the longer route, she drives a total of 42 kilometers.
(2) When she drives both ways, from her home to her office and back, by the longer route, she drives a total of 46 kilometers.

72. Is $x > y$?

(1) $x = y + 2$
(2) $\frac{x}{2} = y - 1$

73. If m is an integer, is m odd?

(1) $\frac{m}{2}$ is not an even integer.
(2) $m - 3$ is an even integer.



74. What is the area of triangular region ABC above?

(1) The product of BD and AC is 20.
(2) $x = 45$

75. In the xy-plane, the line with equation $ax + by + c = 0$, where $abc \neq 0$, has slope $\frac{2}{3}$. What is the value of b?

(1) $a = 4$
(2) $c = -6$

76. If m, p, and t are positive integers and $m < p < t$, is the product mpt an even integer?

(1) $t - p = p - m$
(2) $t - m = 16$

77. Each week a certain salesman is paid a fixed amount equal to \$300, plus a commission equal to 5 percent of the amount of his sales that week over \$1,000. What is the total amount the salesman was paid last week?

(1) The total amount the salesman was paid last week is equal to 10 percent of the amount of his sales last week.
(2) The salesman's sales last week totaled \$5,000.

78. A total of \$60,000 was invested for one year. Part of this amount earned simple annual interest at the rate of x percent per year, and the rest earned simple annual interest at the rate of y percent per year. If the total interest earned by the \$60,000 for that year was \$4,080, what is the value of x?

(1) $x = \frac{3y}{4}$
(2) The ratio of the amount that earned interest at the rate of x percent per year to the amount that earned interest at the rate of y percent per year was 3 to 2.

79. Leo can buy a certain computer for p_1 dollars in State A, where the sales tax is t_1 percent, or he can buy the same computer for p_2 dollars in State B, where the sales tax is t_2 percent. Is the total cost of the computer greater in State A than in State B?

(1) $t_1 > t_2$
(2) $p_1t_1 > p_2t_2$

80. If $r > 0$ and $s > 0$, is $\frac{r}{s} < \frac{s}{r}$?

(1) $\frac{r}{3s} = \frac{1}{4}$
(2) $s = r + 4$

k, n, 12, 6, 17

81. What is the value of n in the list above?

(1) $k < n$
(2) The median of the numbers in the list is 10.

82. If positive integer x is a multiple of 6 and positive integer y is a multiple of 14, is xy a multiple of 105 ?

(1) x is a multiple of 9.
(2) y is a multiple of 25.

83. What is the value of $b + c$?

(1) $ab + cd + ac + bd = 6$
(2) $a + d = 4$

84. What is the average (arithmetic mean) of j and k ?

(1) The average (arithmetic mean) of $j + 2$ and $k + 4$ is 11.
(2) The average (arithmetic mean) of j, k, and 14 is 10.

85. Paula and Sandy were among those people who sold raffle tickets to raise money for Club X. If Paula and Sandy sold a total of 100 of the tickets, how many of the tickets did Paula sell?

(1) Sandy sold $\frac{2}{3}$ as many of the raffle tickets as Paula did.
(2) Sandy sold 8 percent of all the raffle tickets sold for Club X.

86. A number of people each wrote down one of the first 30 positive integers. Were any of the integers written down by more than one of the people?

(1) The number of people who wrote down an integer was greater than 40.
(2) The number of people who wrote down an integer was less than 70.

87. Is the number of seconds required to travel d_1 feet at r_1 feet per second greater than the number of seconds required to travel d_2 feet at r_2 feet per second?

(1) d_1 is 30 greater than d_2.
(2) r_1 is 30 greater than r_2.

88. Last year, if Arturo spent a total of \$12,000 on his mortgage payments, real estate taxes, and home insurance, how much did he spend on his real estate taxes?

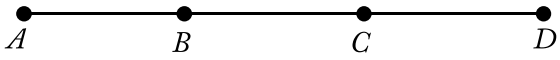
(1) Last year, the total amount that Arturo spent on his real estate taxes and home insurance was $33\frac{1}{3}$ percent of the amount that he spent on his mortgage payments.
(2) Last year, the amount that Arturo spent on his real estate taxes was 20 percent of the total amount he spent on his mortgage payments and home insurance.

89. Is the number of members of Club X greater than the number of members of Club Y ?

(1) Of the members of Club X, 20 percent are also members of Club Y.
(2) Of the members of Club Y, 30 percent are also members of Club X.

90. If k, m, and t are positive integers and $\frac{k}{6}+\frac{m}{4}=\frac{t}{12}$, do t and 12 have a common factor greater than 1 ?

(1) k is a multiple of 3.
(2) m is a multiple of 3.

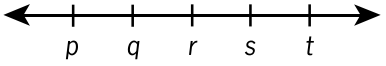


91. In the figure above, is $CD > BC$?

(1) $AD = 20$
(2) $AB = CD$

92. In a certain office, 50 percent of the employees are college graduates and 60 percent of the employees are over 40 years old. If 30 percent of those over 40 have master's degrees, how many of the employees over 40 have master's degrees?

(1) Exactly 100 of the employees are college graduates.
(2) Of the employees 40 years old or less, 25 percent have master's degrees.



93. On the number line above, p, q, r, s, and t are five consecutive even integers in increasing order. What is the average (arithmetic mean) of these five integers?

(1) $q + s = 24$
(2) The average (arithmetic mean) of q and r is 11.

94. If line k in the xy-plane has equation $y = mx + b$, where m and b are constants, what is the slope of k?

(1) k is parallel to the line with equation $y = (1 - m)x + b + 1$.

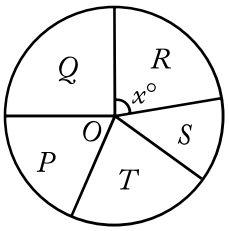
(2) k intersects the line with equation $y = 2x + 3$ at the point (2,7).

95. Is $rst = 1$?

(1) $rs = 1$

(2) $st = 1$

TOTAL EXPENSES FOR THE FIVE DIVISIONS OF COMPANY H



96. The figure above represents a circle graph of Company H's total expenses broken down by the expenses for each of its five divisions. If O is the center of the circle and if Company H's total expenses are \$5,400,000, what are the expenses for Division R?

(1) $x = 94$

(2) The total expenses for Divisions S and T are twice as much as the expenses for Division R.

97. If x is negative, is $x < -3$?

(1) $x^2 > 9$

(2) $x^3 < -9$

98. Seven different numbers are selected from the integers 1 to 100, and each number is divided by 7. What is the sum of the remainders?

(1) The range of the seven remainders is 6.

(2) The seven numbers selected are consecutive integers.

r	s	t
u	v	w
x	y	z

99. Each of the letters in the table above represents one of the numbers 1, 2, or 3, and each of these numbers occurs exactly once in each row and exactly once in each column. What is the value of r?

(1) $v + z = 6$

(2) $s + t + u + x = 6$

100. If $[x]$ denotes the greatest integer less than or equal to x, is $[x] = 0$?

(1) $5x + 1 = 3 + 2x$

(2) $0 < x < 1$

101. Material A costs \$3 per kilogram, and Material B costs \$5 per kilogram. If 10 kilograms of Material K consists of x kilograms of Material A and y kilograms of Material B, is $x > y$?

(1) $y > 4$

(2) The cost of the 10 kilograms of Material K is less than \$40.

102. While on a straight road, Car X and Car Y are traveling at different constant rates. If Car X is now 1 mile ahead of Car Y, how many minutes from now will Car X be 2 miles ahead of Car Y?

(1) Car X is traveling at 50 miles per hour and Car Y is traveling at 40 miles per hour.

(2) Three minutes ago Car X was $\frac{1}{2}$ mile ahead of Car Y.

103. If a certain animated cartoon consists of a total of 17,280 frames on film, how many minutes will it take to run the cartoon?

(1) The cartoon runs without interruption at the rate of 24 frames per second.

(2) It takes 6 times as long to run the cartoon as it takes to rewind the film, and it takes a total of 14 minutes to do both.

104. At what speed was a train traveling on a trip when it had completed half of the total distance of the trip?

(1) The trip was 460 miles long and took 4 hours to complete.

(2) The train traveled at an average rate of 115 miles per hour on the trip.

105. Tom, Jane, and Sue each purchased a new house. The average (arithmetic mean) price of the three houses was $120,000. What was the median price of the three houses?

(1) The price of Tom's house was $110,000.

(2) The price of Jane's house was $120,000.

106. If x and y are integers, is xy even?

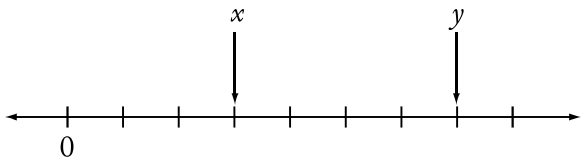
(1) $x = y + 1$

(2) $\frac{x}{y}$ is an even integer.

107. A box contains only red chips, white chips, and blue chips. If a chip is randomly selected from the box, what is the probability that the chip will be either white or blue?

(1) The probability that the chip will be blue is $\frac{1}{5}$.

(2) The probability that the chip will be red is $\frac{1}{3}$.



108. If the successive tick marks shown on the number line above are equally spaced and if x and y are the numbers designating the end points of intervals as shown, what is the value of y?

(1) $x = \frac{1}{2}$

(2) $y - x = \frac{2}{3}$

109. In triangle ABC, point X is the midpoint of side AC and point Y is the midpoint of side BC. If point R is the midpoint of line segment XC and if point S is the midpoint of line segment YC, what is the area of triangular region RCS?

(1) The area of triangular region ABX is 32.

(2) The length of one of the altitudes of triangle ABC is 8.

110. The product of the units digit, the tens digit, and the hundreds digit of the positive integer m is 96. What is the units digit of m?

(1) m is odd.

(2) The hundreds digit of m is 8.

111. A department manager distributed a number of pens, pencils, and pads among the staff in the department, with each staff member receiving x pens, y pencils, and z pads. How many staff members were in the department?

(1) The numbers of pens, pencils, and pads that each staff member received were in the ratio 2:3:4, respectively.

(2) The manager distributed a total of 18 pens, 27 pencils, and 36 pads.

112. Machines X and Y produced identical bottles at different constant rates. Machine X, operating alone for 4 hours, filled part of a production lot; then Machine Y, operating alone for 3 hours, filled the rest of this lot. How many hours would it have taken Machine X operating alone to fill the entire production lot?

(1) Machine X produced 30 bottles per minute.

(2) Machine X produced twice as many bottles in 4 hours as Machine Y produced in 3 hours.

113. On a company-sponsored cruise, $\frac{2}{3}$ of the passengers were company employees and the remaining passengers were their guests. If $\frac{3}{4}$ of the company-employee passengers were managers, what was the number of company-employee passengers who were NOT managers?

(1) There were 690 passengers on the cruise.

(2) There were 230 passengers who were guests of the company employees.

114. The length of the edging that surrounds circular garden K is $\frac{1}{2}$ the length of the edging that surrounds circular garden G. What is the area of garden K? (Assume that the edging has negligible width.)

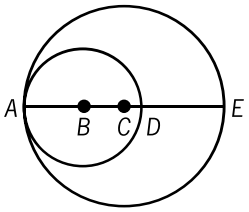
(1) The area of G is 25π square meters.
(2) The edging around G is 10π meters long.

115. For any integers x and y, $\min(x, y)$ and $\max(x, y)$ denote the minimum and the maximum of x and y, respectively. For example, $\min(5, 2) = 2$ and $\max(5, 2) = 5$. For the integer w, what is the value of $\min(10, w)$?

(1) $w = \max(20, z)$ for some integer z.
(2) $w = \max(10, w)$

116. During a 6-day local trade show, the least number of people registered in a single day was 80. Was the average (arithmetic mean) number of people registered per day for the 6 days greater than 90?

(1) For the 4 days with the greatest number of people registered, the average (arithmetic mean) number registered per day was 100.
(2) For the 3 days with the smallest number of people registered, the average (arithmetic mean) number registered per day was 85.



117. In the figure above, points A, B, C, D, and E lie on a line. A is on both circles, B is the center of the smaller circle, C is the center of the larger circle, D is on the smaller circle, and E is on the larger circle. What is the area of the region inside the larger circle and outside the smaller circle?

(1) $AB = 3$ and $BC = 2$
(2) $CD = 1$ and $DE = 4$

118. An employee is paid 1.5 times the regular hourly rate for each hour worked in excess of 40 hours per week, excluding Sunday, and 2 times the regular hourly rate for each hour worked on Sunday. How much was the employee paid last week?

(1) The employee's regular hourly rate is \$10.
(2) Last week the employee worked a total of 54 hours but did not work more than 8 hours on any day.

119. What was the revenue that a theater received from the sale of 400 tickets, some of which were sold at the full price and the remainder of which were sold at a reduced price?

(1) The number of tickets sold at the full price was $\frac{1}{4}$ of the total number of tickets sold.
(2) The full price of a ticket was \$25.

120. The annual rent collected by a corporation from a certain building was x percent more in 1998 than in 1997 and y percent less in 1999 than in 1998. Was the annual rent collected by the corporation from the building more in 1999 than in 1997?

(1) $x > y$
(2) $\frac{xy}{100} < x - y$

121. In the xy-plane, region R consists of all the points (x,y) such that $2x + 3y \leq 6$. Is the point (r,s) in region R?

(1) $3r + 2s = 6$
(2) $r \leq 3$ and $s \leq 2$

122. What is the volume of a certain rectangular solid?

(1) Two adjacent faces of the solid have areas 15 and 24, respectively.
(2) Each of two opposite faces of the solid has area 40.

123. Joanna bought only \$0.15 stamps and \$0.29 stamps. How many \$0.15 stamps did she buy?

(1) She bought \$4.40 worth of stamps.
(2) She bought an equal number of \$0.15 stamps and \$0.29 stamps.

	Favorable	Unfavorable	Not Sure
Candidate M	40	20	40
Candidate N	30	35	35

124. The table above shows the results of a survey of 100 voters who each responded "Favorable" or "Unfavorable" or "Not Sure" when asked about their impressions of Candidate M and of Candidate N. What was the number of voters who responded "Favorable" for both candidates?

(1) The number of voters who did not respond "Favorable" for either candidate was 40.

(2) The number of voters who responded "Unfavorable" for both candidates was 10.

125. If $\circ$ represents one of the operations +, –, and ×, is $k \circ (\ell + m) = (k \circ \ell) + (k \circ m)$ for all numbers k, ℓ, and m?

(1) $k \circ 1$ is not equal to $1 \circ k$ for some numbers k.

(2) $\circ$ represents subtraction.

126. How many of the 60 cars sold last month by a certain dealer had neither power windows nor a stereo?

(1) Of the 60 cars sold, 20 had a stereo but not power windows.

(2) Of the 60 cars sold, 30 had both power windows and a stereo.

127. In Jefferson School, 300 students study French or Spanish or both. If 100 of these students do not study French, how many of these students study both French and Spanish?

(1) Of the 300 students, 60 do not study Spanish.

(2) A total of 240 of the students study Spanish.

128. A school administrator will assign each student in a group of n students to one of m classrooms. If $3 < m < 13 < n$, is it possible to assign each of the n students to one of the m classrooms so that each classroom has the same number of students assigned to it?

(1) It is possible to assign each of $3n$ students to one of m classrooms so that each classroom has the same number of students assigned to it.

(2) It is possible to assign each of $13n$ students to one of m classrooms so that each classroom has the same number of students assigned to it.

129. What is the median number of employees assigned per project for the projects at Company Z?

(1) 25 percent of the projects at Company Z have 4 or more employees assigned to each project.

(2) 35 percent of the projects at Company Z have 2 or fewer employees assigned to each project.

130. If Juan had a doctor's appointment on a certain day, was the appointment on a Wednesday?

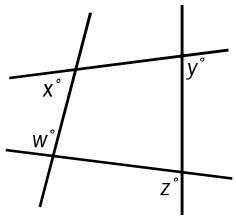
(1) Exactly 60 hours before the appointment, it was Monday.

(2) The appointment was between 1:00 p.m. and 9:00 p.m.

131. When a player in a certain game tossed a coin a number of times, 4 more heads than tails resulted. Heads or tails resulted each time the player tossed the coin. How many times did heads result?

(1) The player tossed the coin 24 times.

(2) The player received 3 points each time heads resulted and 1 point each time tails resulted, for a total of 52 points.



132. What is the value of $x + y$ in the figure above?

(1) $w = 95$

(2) $z = 125$

133. Are all of the numbers in a certain list of 15 numbers equal?

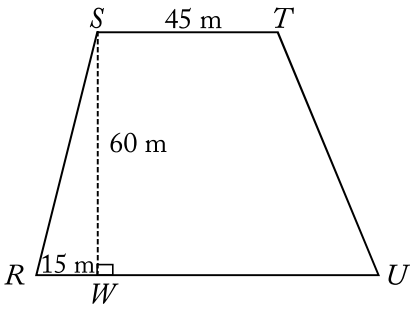
(1) The sum of all the numbers in the list is 60.

(2) The sum of any 3 numbers in the list is 12.

134. A scientist recorded the number of eggs in each of 10 birds' nests. What was the standard deviation of the numbers of eggs in the 10 nests?

(1) The average (arithmetic mean) number of eggs for the 10 nests was 4.

(2) Each of the 10 nests contained the same number of eggs.



135. Quadrilateral *RSTU* shown above is a site plan for a parking lot in which side *RU* is parallel to side *ST* and *RU* is longer than *ST*. What is the area of the parking lot?

(1) *RU* = 80 meters

(2) *TU* = $20\sqrt{10}$ meters

136. If the average (arithmetic mean) of six numbers is 75, how many of the numbers are equal to 75 ?

(1) None of the six numbers is less than 75.

(2) None of the six numbers is greater than 75.

137. At a bakery, all donuts are priced equally and all bagels are priced equally. What is the total price of 5 donuts and 3 bagels at the bakery?

(1) At the bakery, the total price of 10 donuts and 6 bagels is \$12.90.

(2) At the bakery, the price of a donut is \$0.15 less than the price of a bagel.

138. What was the total amount of revenue that a theater received from the sale of 400 tickets, some of which were sold at *x* percent of full price and the rest of which were sold at full price?

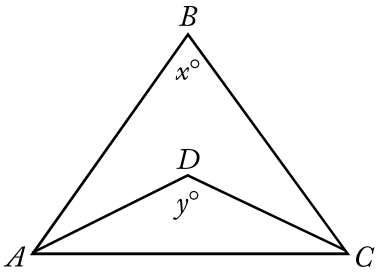
(1) $x = 50$

(2) Full-price tickets sold for \$20 each.

139. Any decimal that has only a finite number of nonzero digits is a terminating decimal. For example, 24, 0.82, and 5.096 are three terminating decimals. If *r* and *s* are positive integers and the ratio $\frac{r}{s}$ is expressed as a decimal, is $\frac{r}{s}$ a terminating decimal?

(1) $90 < r < 100$

(2) $s = 4$



140. In the figure above, what is the value of $x + y$?

(1) $x = 70$

(2) $\triangle ABC$ and $\triangle ADC$ are both isosceles triangles.

141. Committee X and Committee Y, which have no common members, will combine to form Committee Z. Does Committee X have more members than Committee Y ?

(1) The average (arithmetic mean) age of the members of Committee X is 25.7 years and the average age of the members of Committee Y is 29.3 years.

(2) The average (arithmetic mean) age of the members of Committee Z will be 26.6 years.

142. What amount did Jean earn from the commission on her sales in the first half of 1988 ?

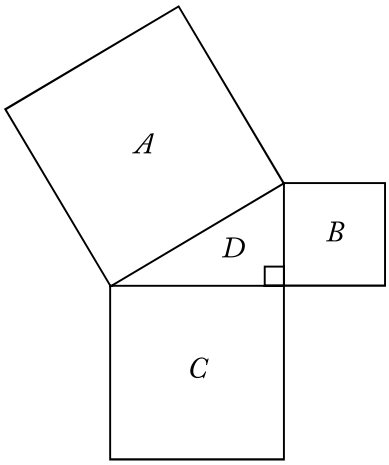
(1) In 1988 Jean's commission was 5 percent of the total amount of her sales.

(2) The amount of Jean's sales in the second half of 1988 averaged \$10,000 per month more than in the first half.

143. The price per share of Stock X increased by 10 percent over the same time period that the price per share of Stock Y decreased by 10 percent. The reduced price per share of Stock Y was what percent of the original price per share of Stock X ?

(1) The increased price per share of Stock X was equal to the original price per share of Stock Y.

(2) The increase in the price per share of Stock X was $\frac{10}{11}$ the decrease in the price per share of Stock Y.



144. In the figure above, if the area of triangular region D is 4, what is the length of a side of square region A?

(1) The area of square region B is 9.

(2) The area of square region C is $\frac{64}{9}$.

145. If Sara's age is exactly twice Bill's age, what is Sara's age?

(1) Four years ago, Sara's age was exactly 3 times Bill's age.

(2) Eight years from now, Sara's age will be exactly 1.5 times Bill's age.

146. A report consisting of 2,600 words is divided into 23 paragraphs. A 2-paragraph preface is then added to the report. Is the average (arithmetic mean) number of words per paragraph for all 25 paragraphs less than 120?

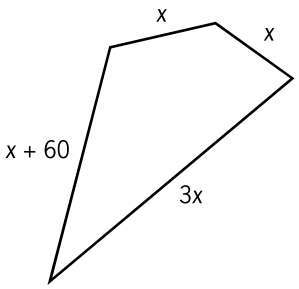
(1) Each paragraph of the preface has more than 100 words.

(2) Each paragraph of the preface has fewer than 150 words.

147. A certain bookcase has 2 shelves of books. On the upper shelf, the book with the greatest number of pages has 400 pages. On the lower shelf, the book with the least number of pages has 475 pages. What is the median number of pages for all of the books on the 2 shelves?

(1) There are 25 books on the upper shelf.

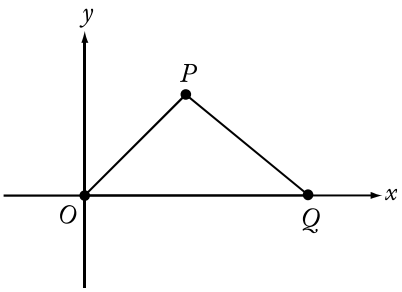
(2) There are 24 books on the lower shelf.



148. The figure above shows the number of meters in the lengths of the four sides of a jogging path. What is the total distance around the path?

(1) One of the sides of the path is 120 meters long.

(2) One of the sides of the path is twice as long as each of the two shortest sides.



149. In the rectangular coordinate system above, if $OP < PQ$, is the area of region OPQ greater than 48?

(1) The coordinates of point P are (6,8).

(2) The coordinates of point Q are (13,0).

$$S = \frac{\frac{2}{n}}{\frac{1}{x} + \frac{2}{3x}}$$

150. In the expression above, if $xn \neq 0$, what is the value of S?

(1) $x = 2n$

(2) $n = \frac{1}{2}$

151. If n is a positive integer and $k = 5.1 \times 10^n$, what is the value of k?

(1) $6{,}000 < k < 500{,}000$

(2) $k^2 = 2.601 \times 10^9$

152. If Carmen had 12 more tapes, she would have twice as many tapes as Rafael. Does Carmen have fewer tapes than Rafael?

(1) Rafael has more than 5 tapes.
(2) Carmen has fewer than 12 tapes.

153. If x is an integer, is $x|x| < 2^x$?

(1) $x < 0$
(2) $x = -10$

154. If n is a positive integer, is the value of $b - a$ at least twice the value of $3^n - 2^n$?

(1) $a = 2^{n+1}$ and $b = 3^{n+1}$
(2) $n = 3$

155. The inflation index for the year 1989 relative to the year 1970 was 3.56, indicating that, on the average, for each dollar spent in 1970 for goods, $3.56 had to be spent for the same goods in 1989. If the price of a Model K mixer increased precisely according to the inflation index, what was the price of the mixer in 1970?

(1) The price of the Model K mixer was $102.40 more in 1989 than in 1970.
(2) The price of the Model K mixer was $142.40 in 1989.

156. Is 5^k less than 1,000?

(1) $5^{k+1} > 3{,}000$
(2) $5^{k-1} = 5^k - 500$

157. The hypotenuse of a right triangle is 10 cm. What is the perimeter, in centimeters, of the triangle?

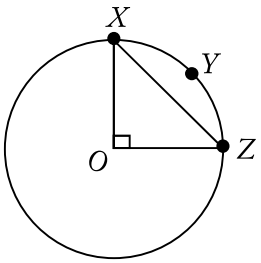
(1) The area of the triangle is 25 square centimeters.
(2) The 2 legs of the triangle are of equal length.

158. Every member of a certain club volunteers to contribute equally to the purchase of a $60 gift certificate. How many members does the club have?

(1) Each member's contribution is to be $4.
(2) If 5 club members fail to contribute, the share of each contributing member will increase by $2.

159. If $x < 0$, is $y > 0$?

(1) $\frac{x}{y} < 0$
(2) $y - x > 0$



160. What is the circumference of the circle above with center O?

(1) The perimeter of ΔOXZ is $20 + 10\sqrt{2}$.
(2) The length of arc XYZ is 5π.

161. Beginning in January of last year, Carl made deposits of $120 into his account on the 15th of each month for several consecutive months and then made withdrawals of $50 from the account on the 15th of each of the remaining months of last year. There were no other transactions in the account last year. If the closing balance of Carl's account for May of last year was $2,600, what was the range of the monthly closing balances of Carl's account last year?

(1) Last year the closing balance of Carl's account for April was less than $2,625.
(2) Last year the closing balance of Carl's account for June was less than $2,675.

162. If n and k are positive integers, is $\sqrt{n+k} > 2\sqrt{n}$?

(1) $k > 3n$
(2) $n + k > 3n$

163. In a certain business, production index p is directly proportional to efficiency index e, which is in turn directly proportional to investment index i. What is p if $i = 70$?

(1) $e = 0.5$ whenever $i = 60$.
(2) $p = 2.0$ whenever $i = 50$.

164. In the rectangular coordinate system, are the points (r,s) and (u,v) equidistant from the origin?

(1) $r + s = 1$

(2) $u = 1 - r$ and $v = 1 - s$

165. If x is an integer, is $9^x + 9^{-x} = b$?

(1) $3^x + 3^{-x} = \sqrt{b+2}$

(2) $x > 0$

166. If n is a positive integer, is $\left(\frac{1}{10}\right)^n < 0.01$?

(1) $n > 2$

(2) $\left(\frac{1}{10}\right)^{n-1} < 0.1$

167. If n is a positive integer, what is the tens digit of n ?

(1) The hundreds digit of $10n$ is 6.

(2) The tens digit of $n + 1$ is 7.

168. What is the value of $\frac{2t+t-x}{t-x}$?

(1) $\frac{2t}{t-x} = 3$

(2) $t - x = 5$

169. Is n an integer?

(1) n^2 is an integer.

(2) $\sqrt{n}$ is an integer.

170. If n is a positive integer, is $n^3 - n$ divisible by 4 ?

(1) $n = 2k + 1$, where k is an integer.

(2) $n^2 + n$ is divisible by 6.

171. What is the tens digit of positive integer x ?

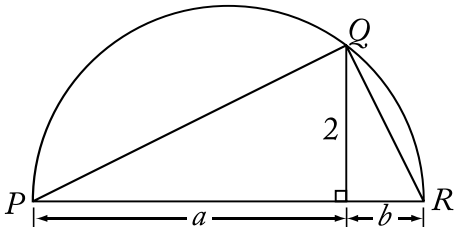
(1) x divided by 100 has a remainder of 30.

(2) x divided by 110 has a remainder of 30.

172. If x, y, and z are positive integers, is $x - y$ odd?

(1) $x = z^2$

(2) $y = (z - 1)^2$



173. If arc PQR above is a semicircle, what is the length of diameter $\overline{PR}$?

(1) $a = 4$

(2) $b = 1$

174. Marcia's bucket can hold a maximum of how many liters of water?

(1) The bucket currently contains 9 liters of water.

(2) If 3 liters of water are added to the bucket when it is half full of water, the amount of water in the bucket will increase by $\frac{1}{3}$.